

FORTINET


KENT STATE[®]
UNIVERSITY

Kent State University's Border Firewall Architecture Design with Fortinet

OHECC 2024

- Presented by:
 - Tom Bordonaro – Network Nerd
 - AJ Chiccarino – Security Dork
 - Victor Weis – Fortinet Renaissance Man

Introductions

- **Tom Bordonaro**
 - Network Engineer
- **AJ Chiccarino**
 - Senior Manager – Security Engineering
- **About Kent State**
 - 33,000 Students
 - 3rd Largest University in Ohio
 - 8 Regional campuses
 - 150 + Buildings
 - 1100+ Switches
 - 5600+ Wireless Access Points
 - 20 FortiGate Firewalls

Introductions

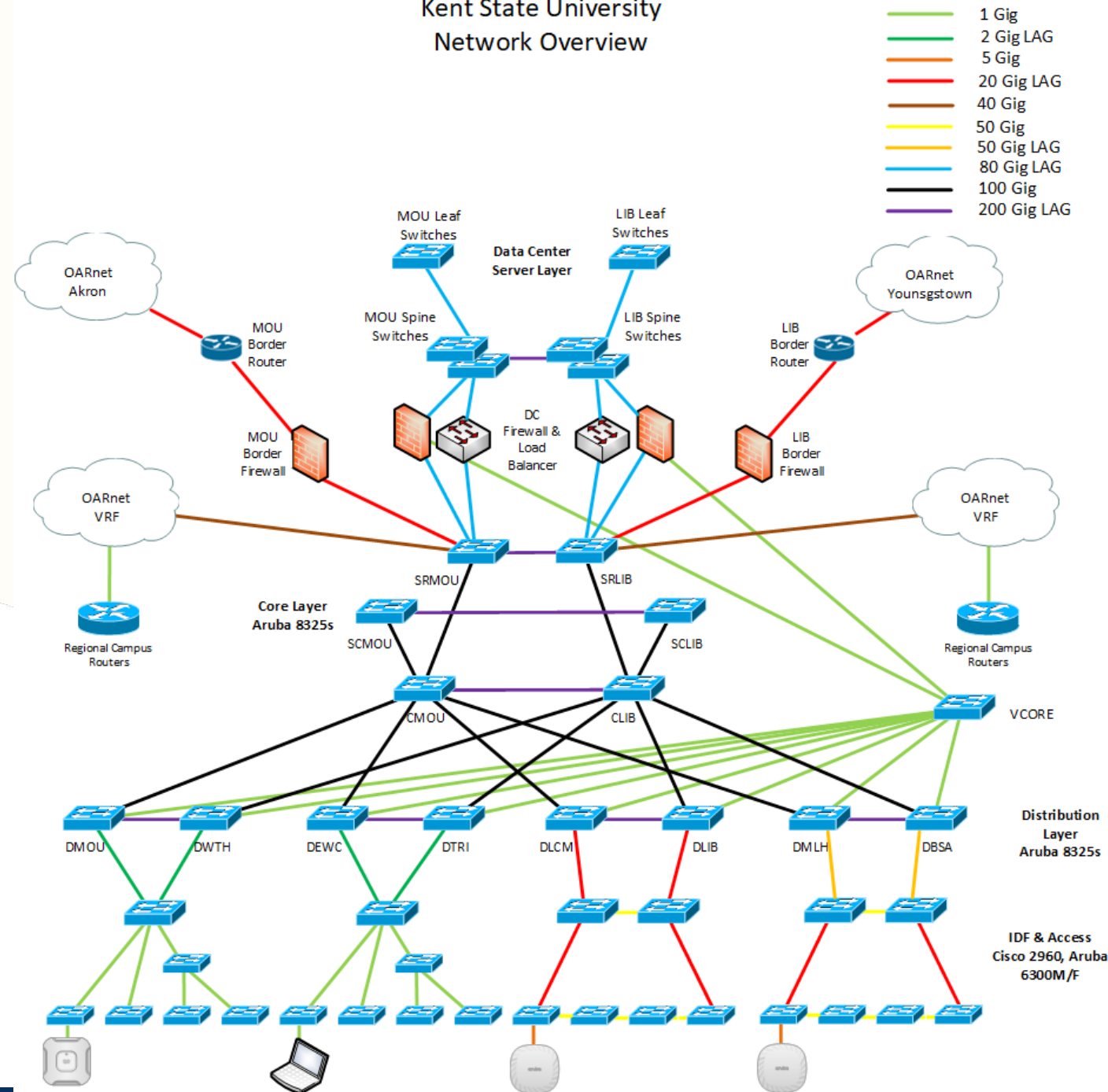
- **Victor Weis**
 - Consulting Solution Architect
 - Lives in Dayton, OH – born & raised
- **About Fortinet**
 - Cybersecurity, everywhere you need it.

Prior to May 2023

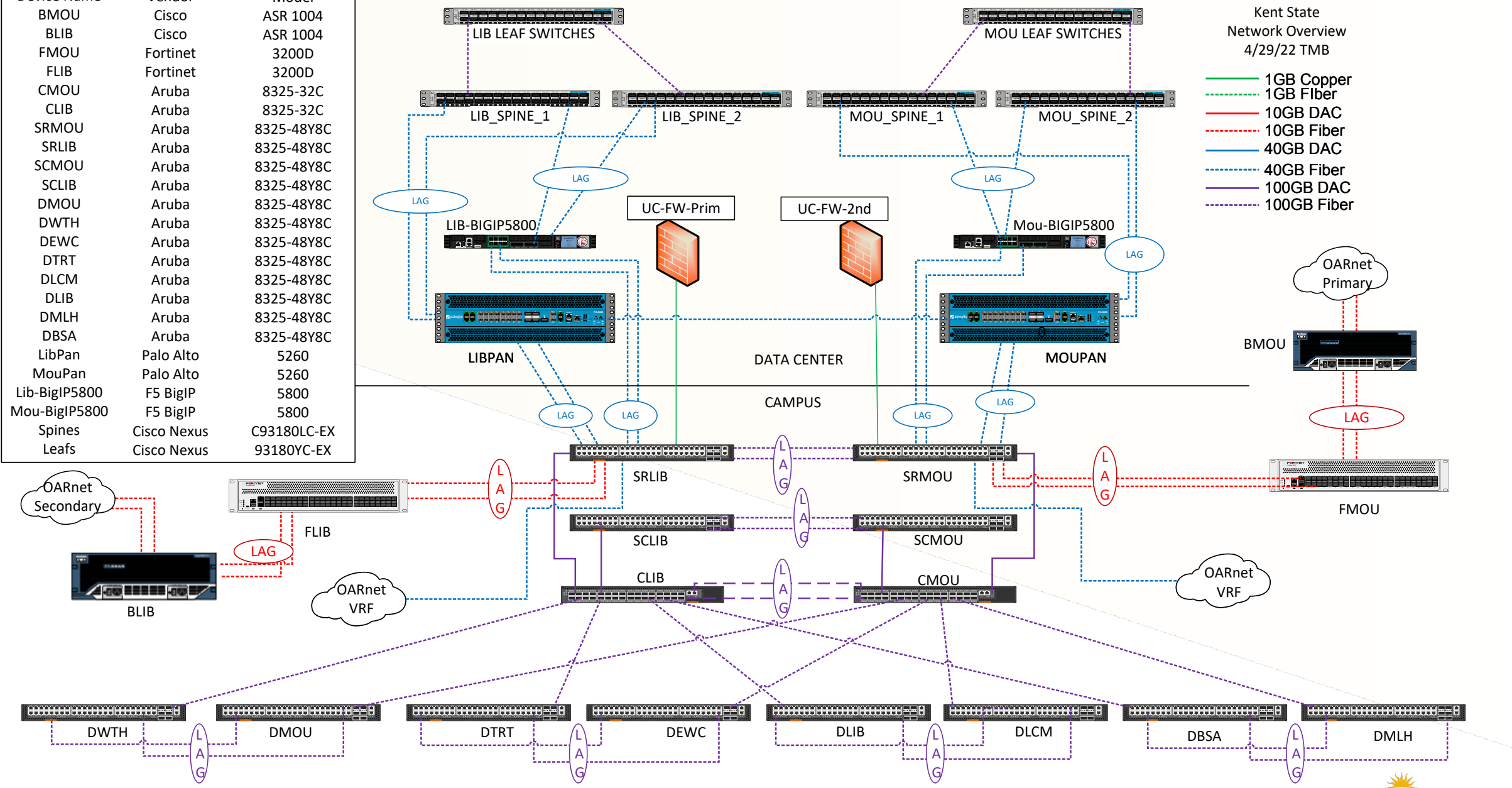
Why we needed to change?

- 3200Ds (2017) with FGSP HA session sync (FGSP)
- Active – active – complexity
- Throughput/performance
- One default gateway
- Memory conserve mode - DDOS
- Shaping
- Regional Backup VPNs
- Cisco ASR Smartnet \$

Kent State University
Network Overview

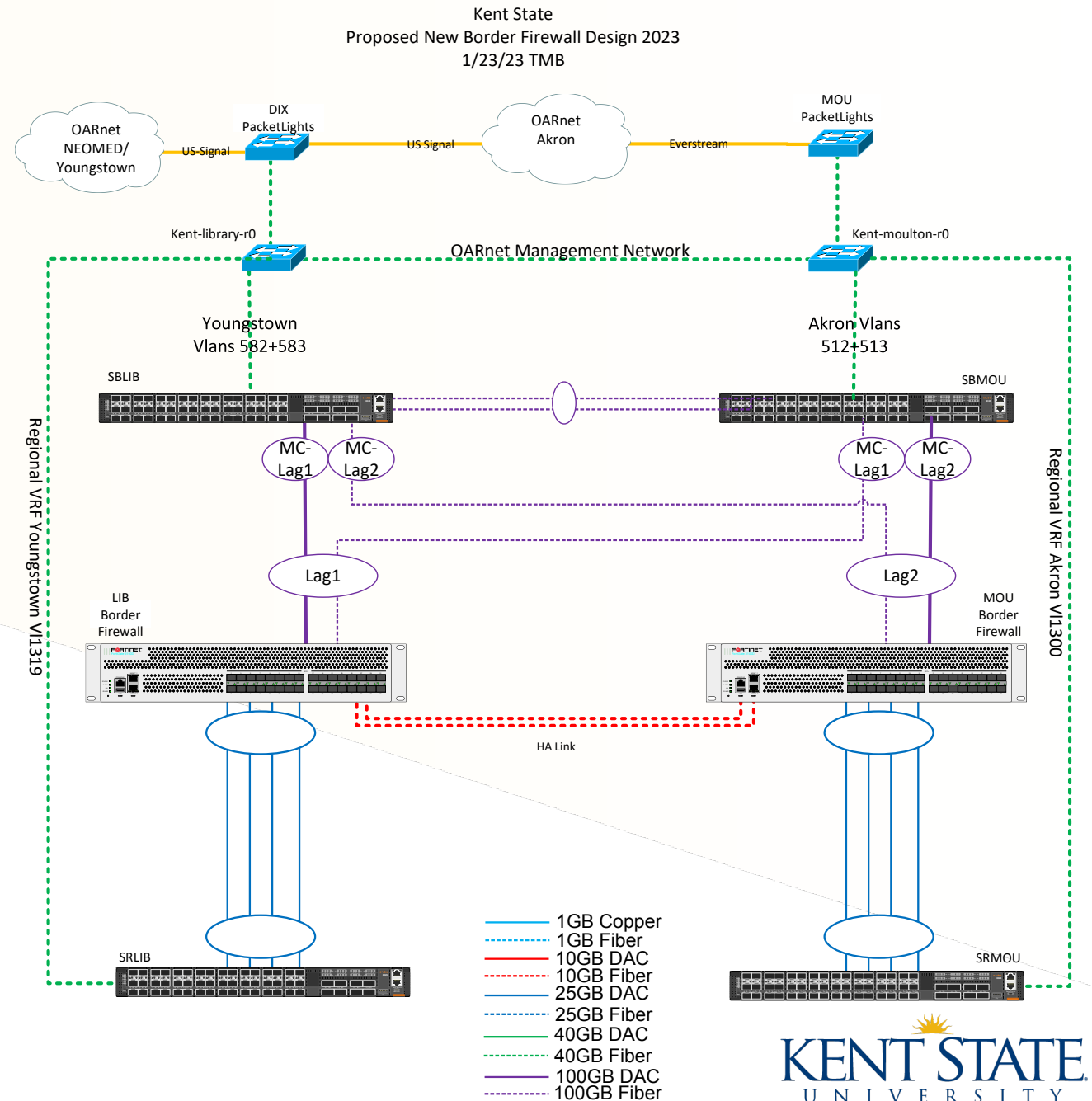


Device Name	Vendor	Model
BMOU	Cisco	ASR 1004
BLIB	Cisco	ASR 1004
FMOU	Fortinet	3200D
FLIB	Fortinet	3200D
CMOU	Aruba	8325-32C
CLIB	Aruba	8325-32C
SRMOU	Aruba	8325-48Y8C
SRLIB	Aruba	8325-48Y8C
SCMOU	Aruba	8325-48Y8C
SCLIB	Aruba	8325-48Y8C
DMOU	Aruba	8325-48Y8C
DWTH	Aruba	8325-48Y8C
DEWC	Aruba	8325-48Y8C
DTRT	Aruba	8325-48Y8C
DLCM	Aruba	8325-48Y8C
DLIB	Aruba	8325-48Y8C
DMLH	Aruba	8325-48Y8C
DBSA	Aruba	8325-48Y8C
LibPan	Palo Alto	5260
MouPan	Palo Alto	5260
Lib-BigIP5800	F5 BigIP	5800
Mou-BigIP5800	F5 BigIP	5800
Spines	Cisco Nexus	C93180LC-EX
Leafs	Cisco Nexus	93180YC-EX



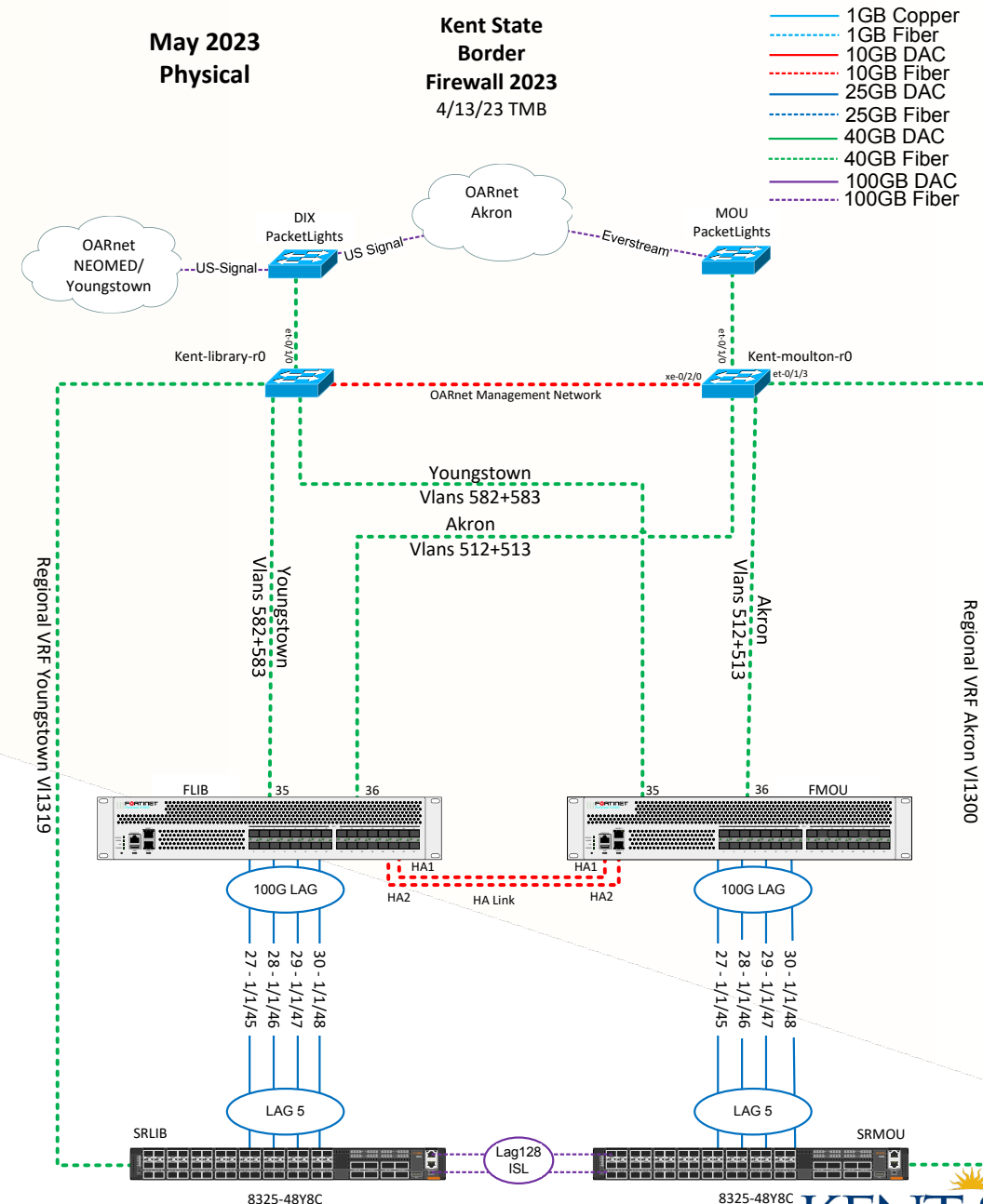
New Design Concept

- Fortigate 3501Fs
- HA Mode:FGCP (active-passive)
- Eliminate Border Routers
- OARnet – Outside Switching Layer
- Create VDOMs – asymmetric traffic allowed
- Eliminate most shaping



New Physical Design

- Physical Port Limitation
- Testbed
- Testing, tweaking, testing
- Production May 2023

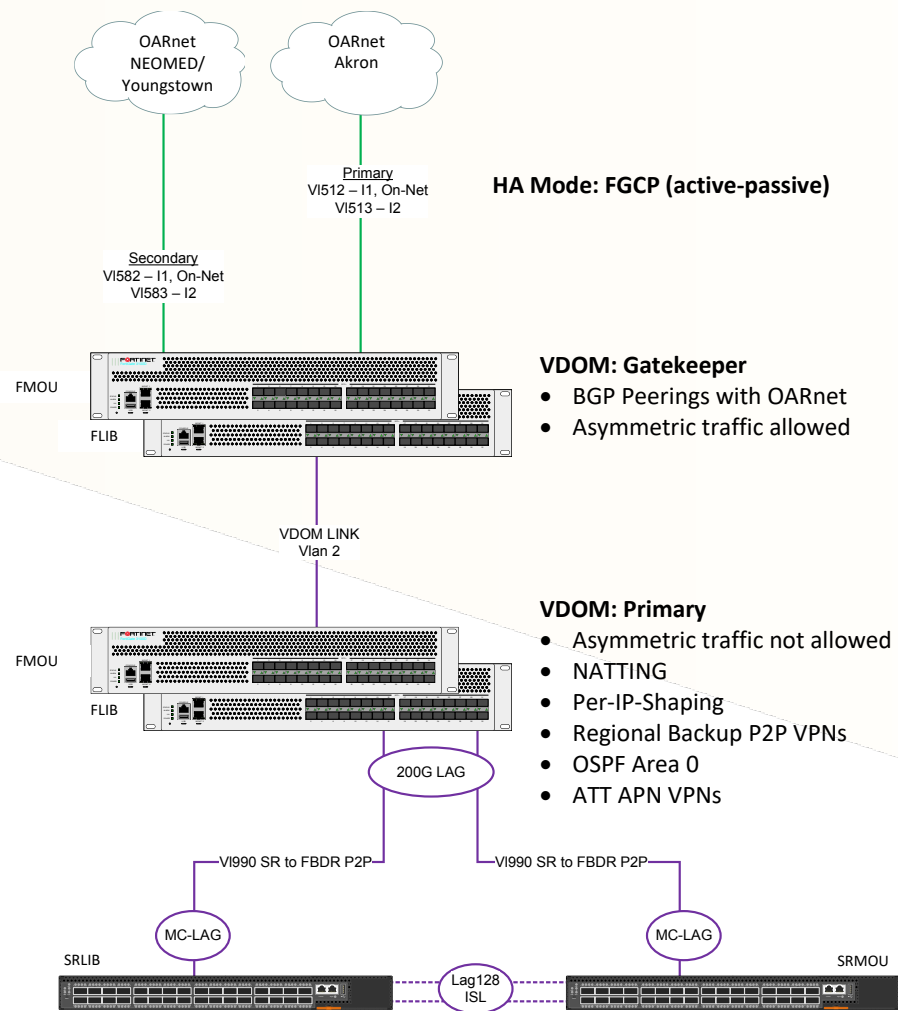


Logical

Logical

Kent State
Border
Firewall
3/25/24 TMB

1GB
10GB
25GB
40GB
100GB

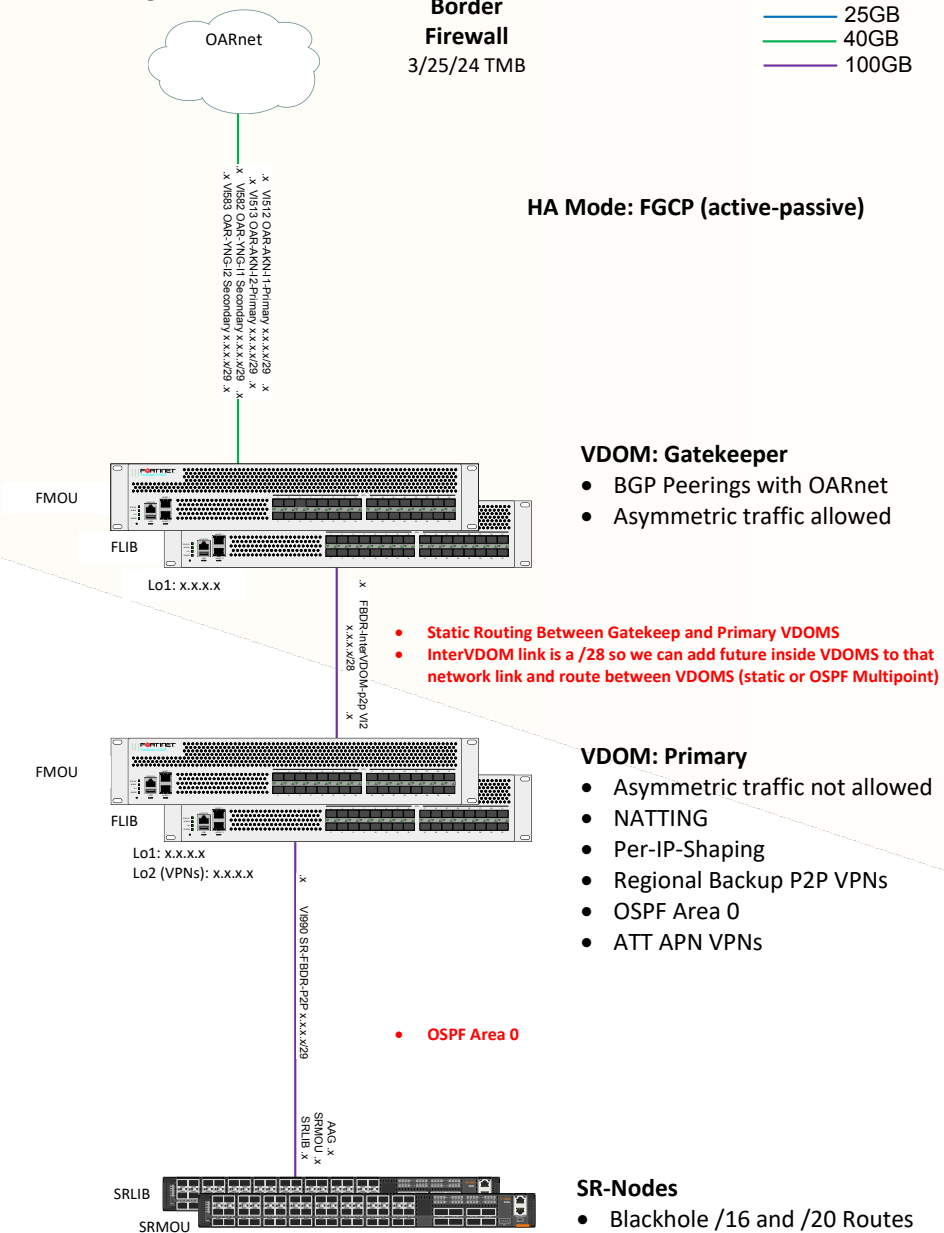


Routing

L3 - Routing

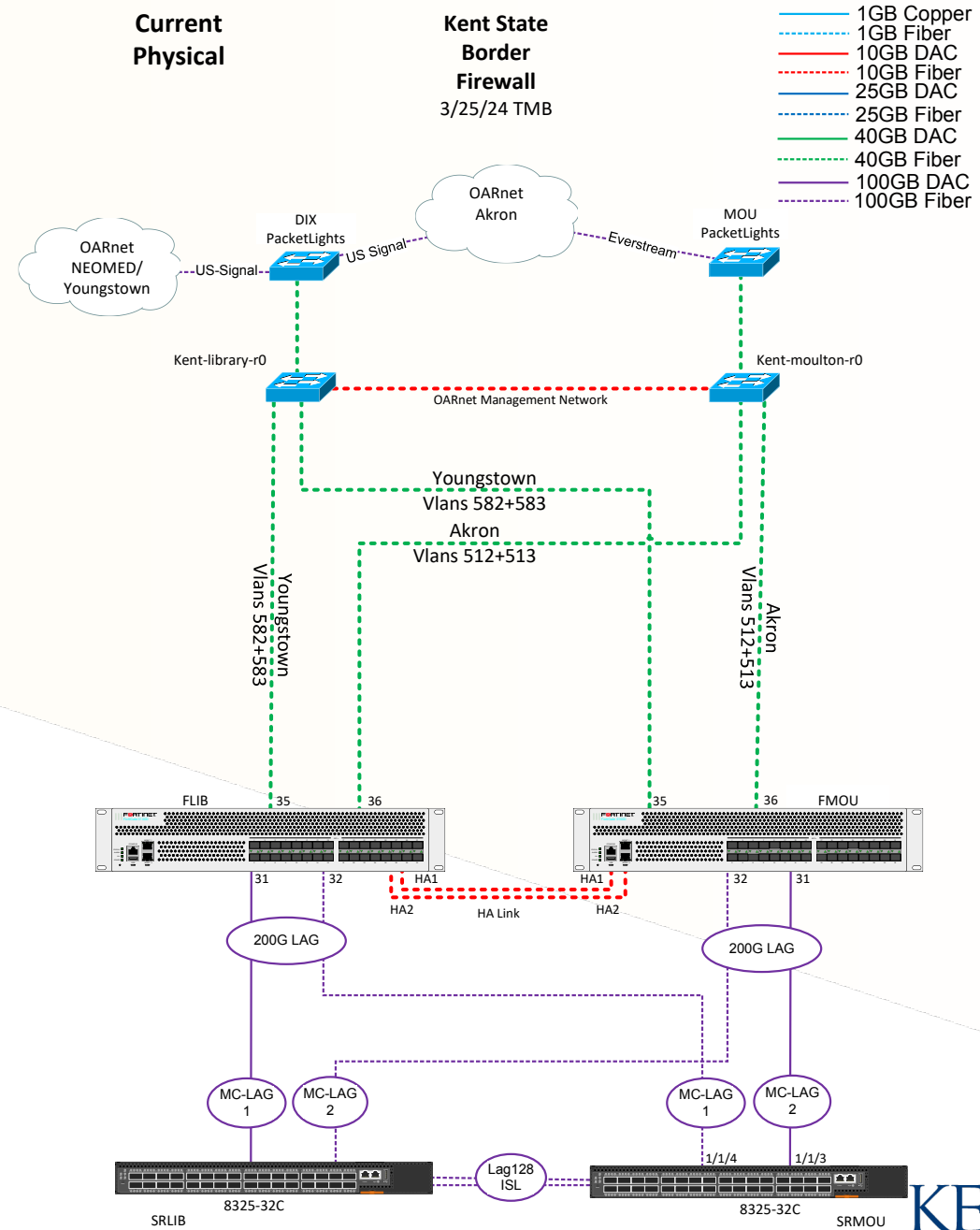
Kent State
Border
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Changes Made in May 2024

- SRnodes 8325-48Y8C to 8325-32C
- 100Gbps LAG to 200Gbps MCLAG



Benefits

- **Session/second count increase**
- **Simplification**
- **Upgrades easier**

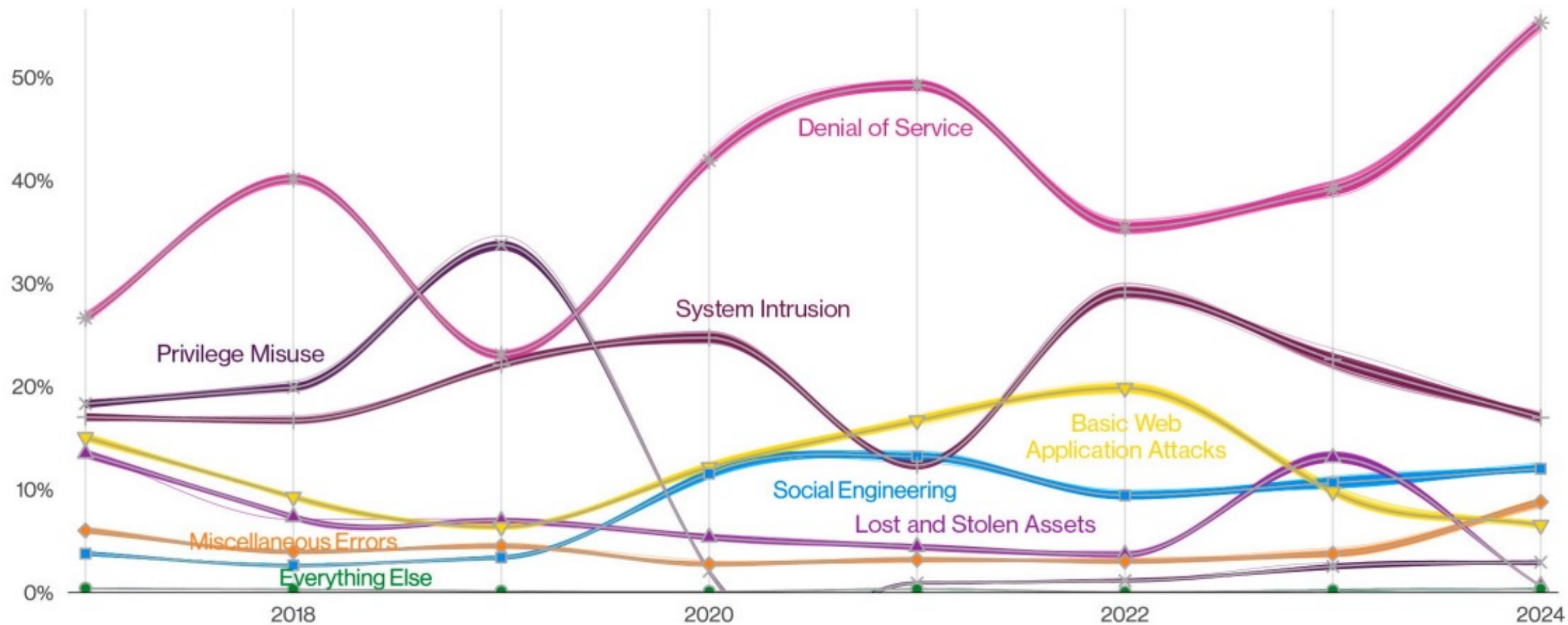


Figure 26. Patterns over time in incidents

Lessons Learned

- **FortiAnalyzer Logging Limits**
 - Multi-Vdom logging replication
- **FortiConverter Service**
 - 3501F to 500E
- **Buy the most capable firewall you can afford**

Future Upgrades/Designs

- **New VDOMs**
- **Regional Backup VPNs**

Questions?

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- AJ Chiccarino – achiccar@kent.edu
- Victor Weis - vweis@fortinet.com